EXPERIMENT 07

Part-of-Speech Tagging Using NLTK

Aim

To tag words with their parts of speech using the NLTK library.

Procedure

1. Install NLTK if required.
2. Import word tokenization and POS-tagging functions.
3. Tokenize the sample sentence.
4. Run the POS tagger.
5. Display the tagged words.

Program

```
# Install first: pip install nltk
import nltk
from nltk import word_tokenize, pos_tag

# Run once if resources are missing:
# nltk.download("punkt")
# nltk.download("punkt_tab")
# nltk.download("averaged_perceptron_tagger_eng")

sentence = "The small cat sleeps peacefully."
tokens = word_tokenize(sentence)
print(pos_tag(tokens))
```

Output

```
OUTPUT — EXPERIMENT 07
[('The', 'DT'), ('small', 'JJ'), ('cat', 'NN'),
 ('sleeps', 'VBZ'), ('peacefully', 'RB'), ('.', '.')]
(Exact tags can vary with NLTK model versions.)
```

Result

Thus, NLTK assigned part-of-speech tags to the words in the sentence.